

1. (12 points) Find the equation for the plane that passes through the point $(1, 2, 4)$ and contains the line given by $x = 2t$, $y = 3 - t$, $z = 2 + 5t$. And give the x -intercept of this plane.

Plane contains $(1, 2, 4)$ & $(0, 3, 2)$: contains vector $\langle -1, 1, -2 \rangle$

Also contains dir. of line: $\langle 2, -1, 5 \rangle$.

Normal vector is orth. to both: $\langle -1, 1, -2 \rangle \times \langle 2, -1, 5 \rangle = \langle 3, 1, -1 \rangle$

Plane $\perp$ normal vector $\langle 3, 1, -1 \rangle$ through $(1, 2, 4)$:

$$3(x-1) + (y-2) - (z-4) = 0$$

$$3x + y - z = 1$$

x -intercept: $\downarrow y=z=0$

$$3x = 1 \rightarrow x = \frac{1}{3}$$

Plane Equation:

$$3x + y - z = 1$$

x -intercept: $(x, y, z) =$

$$\left(\frac{1}{3}, 0, 0 \right)$$

2. (12 points) Let a curve γ be the intersection of the plane $y+z=1$ and the cylinder $x^2+y^2=1$. Find the points where γ has maximum curvature.

$$x = \cos t$$

$$y = \sin t$$

$$z = 1 - y = 1 - \sin t$$

$$\vec{r}(t) = \langle \cos t, \sin t, 1 - \sin t \rangle$$

$$\vec{r}'(t) = \langle -\sin t, \cos t, -\cos t \rangle$$

$$\vec{r}''(t) = \langle -\cos t, -\sin t, \sin t \rangle$$

$$\vec{r}'(t) \times \vec{r}''(t) = \langle 0, 1, 1 \rangle$$

$$|\vec{r}'(t)| = \sqrt{\sin^2 t + \cos^2 t + \cos^2 t} = \sqrt{1 + \cos^2 t}$$

$$|\vec{r}'(t) \times \vec{r}''(t)| = \sqrt{2}$$

$$K = \frac{\sqrt{2}}{(1 + \cos^2 t)^{3/2}} \quad \text{largest when denom. is smallest} \rightarrow \cos^2 t = 0 \rightarrow t = \frac{\pi}{2} \text{ or } \frac{3\pi}{2}$$

$$\vec{r}\left(\frac{\pi}{2}\right) = \langle 0, 1, 0 \rangle$$

$$\vec{r}\left(\frac{3\pi}{2}\right) = \langle 0, -1, 2 \rangle$$

Points: $(x, y, z) =$

$$(0, 1, 0) \text{ \& } (0, -1, 2)$$

3. (6 points per part) Parts (a) and (b), consider the surface implicitly defined by the equation

$$z(\cos(z) + 2) = x^2 - y^2.$$

(a) Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.

$$\frac{\partial z}{\partial x}(\cos z + 2) - z \sin z \frac{\partial z}{\partial x} = 2x$$

$$\frac{\partial z}{\partial x}(\cos z + 2 - z \sin z) = 2x$$

$$\frac{\partial z}{\partial y}(\cos z + 2) - z \sin z \frac{\partial z}{\partial y} = -2y$$

$$\frac{\partial z}{\partial y}(\cos z + 2 - z \sin z) = -2y$$

$$\frac{\partial z}{\partial x} = \frac{2x}{\cos z + 2 - z \sin z}$$

$$\frac{\partial z}{\partial y} = \frac{-2y}{\cos z + 2 - z \sin z}$$

- (b) Give the linear approximation, i.e. tangent plane, to the surface at the point where $(x, y) = (1, 1)$. And use this approximation to estimate the z -coordinate of a point on the surface where $(x, y) = (1.01, 0.98)$.

When $x=y=1 \rightarrow z(\cos z + 2) = 0 \rightarrow z=0$ ($\cos z + 2$ is never 0)

$$\frac{\partial z}{\partial x} = \frac{2}{1+2} = \frac{2}{3} \quad \frac{\partial z}{\partial y} = \frac{-2}{1+2} = -\frac{2}{3}$$

$$L(x, y) = \frac{2}{3}(x-1) - \frac{2}{3}(y-1) + 0$$

$$L(1.01, 0.98) = \frac{2}{3}(0.01) - \frac{2}{3}(-0.02) = 0.02$$

Linearization: $L(x, y) = \frac{2}{3}(x-1) - \frac{2}{3}(y-1)$

$$z \approx 0.02$$

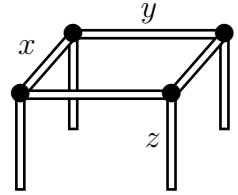
4. (12 points) Find the dimensions of a rectangular tent of maximum volume such that the sum of the lengths of its 8 posts (above the ground) is 36 m.

To receive full credit, you must justify that your answer is indeed the maximum.

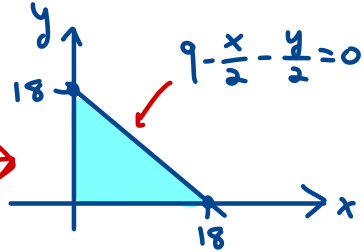
Want to maximize $V = xyz$

Constraint: $2x + 2y + 4z = 36 \rightarrow z = 9 - \frac{x}{2} - \frac{y}{2}$

Let $f(x, y) = V = xy(9 - \frac{x}{2} - \frac{y}{2}) = 9xy - \frac{1}{2}x^2y - \frac{1}{2}xy^2$



Domain: $x \geq 0, y \geq 0, z = 9 - \frac{x}{2} - \frac{y}{2} \geq 0$:



Note: on boundary, x, y , or z is zero, so

volume = 0. So max cannot be on boundary! Must be at crit pt.

$$\begin{aligned} f_x(x, y) &= 9y - xy - \frac{1}{2}y^2 = 0 \\ f_y(x, y) &= 9x - \frac{1}{2}x^2 - xy = 0 \end{aligned} \quad \left. \begin{array}{l} \\ \end{array} \right\} \begin{array}{l} \text{since } x \neq 0 \text{ and } y \neq 0 \text{ at max} \\ \text{we can safely divide by } x \text{ or } y \end{array}$$

$$9 - x - \frac{1}{2}y = 0$$

$$-2 \left(9 - \frac{1}{2}x - y = 0 \right)$$

$$-9 + \frac{3}{2}y = 0$$

$$y = 6 \rightarrow 9 - x - 3 = 0 \rightarrow x = 6$$

$$z = 9 - 3 - 3 = 3$$

This is the only crit pt. not on boundary
so it's the max.

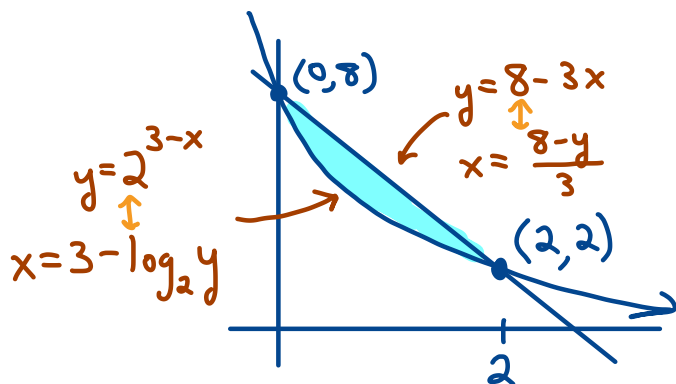
$$x = 6 \text{ m} \quad y = 6 \text{ m} \quad z = 3 \text{ m}$$

Dimensions:

5. (6 points per part) For the following problems, set up an iterated integral as described.

You do not have to evaluate the integrals. Just set them up.

- (a) Reverse the order of integration of $\int_0^2 \int_{2^{3-x}}^{8-3x} \sin(x^2) dy dx$.



Integral: _____

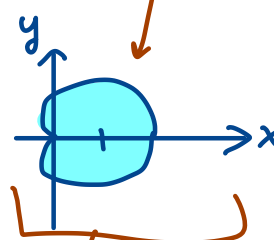
$$\int_2^8 \int_{3-\log_2 y}^{\frac{8-y}{3}} \sin(x^2) dx dy$$

- (b) Write a double integral for the volume below the cone $z = \sqrt{x^2 + y^2}$ and above the paraboloid $z = x^2 - x + y^2$.

(Hint: convert to polar!)

Intersection: $\sqrt{x^2 + y^2} = x^2 - x + y^2$ $\xrightarrow{\text{polar}}$ $r = r^2 - r \cos \theta \rightarrow r = 1 + \cos \theta$

$$\begin{aligned} \text{Height} &= \sqrt{x^2 + y^2} - (x^2 - x + y^2) \\ &= r - r^2 + r \cos \theta \end{aligned}$$

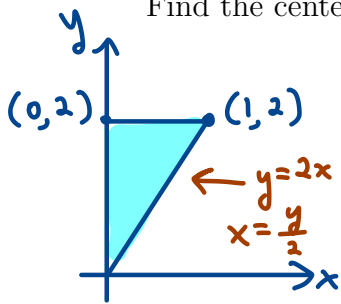


Integral: _____

$$\int_0^{2\pi} \int_0^{1+\cos \theta} (r - r^2 + r \cos \theta) r dr d\theta$$

6. (12 points) Let $\mathcal{L}$ be the lamina whose shape is the triangular region with vertices $(0, 0)$, $(0, 2)$, and $(1, 2)$, and whose density at any point is proportional to its distance to the y -axis.

Find the center of mass of $\mathcal{L}$.



$$\rho(x, y) = k|x|$$

(can just use $\rho(x, y) = x$ to find C.o.M. here)

$$\text{Mass} \quad m = \int_0^2 \int_0^{y/2} x \, dx \, dy = \int_0^2 \left(\frac{1}{2} x^2 \right) \Big|_{x=0}^{x=y/2} dy = \int_0^2 \frac{y^2}{8} dy = \left[\frac{y^3}{24} \right]_0^2 = \boxed{\frac{1}{3}}$$

$$\text{Moments} \quad M_y = \int_0^2 \int_0^{y/2} x^2 \, dx \, dy = \int_0^2 \left(\frac{1}{3} x^3 \right) \Big|_{x=0}^{x=y/2} dy = \int_0^2 \frac{y^3}{24} dy = \left(\frac{y^4}{96} \right) \Big|_0^2 = \boxed{\frac{1}{6}}$$

$$M_x = \int_0^2 \int_0^{y/2} xy \, dx \, dy = \int_0^2 \left(\frac{1}{2} x^2 y \right) \Big|_{x=0}^{x=y/2} dy = \int_0^2 \frac{y^3}{8} dy = \left[\frac{y^4}{32} \right]_0^2 = \boxed{\frac{1}{2}}$$

$$\text{Center of mass} = \left(\frac{M_y}{m}, \frac{M_x}{m} \right) = \left(\frac{\frac{1}{6}}{\frac{1}{3}}, \frac{\frac{1}{2}}{\frac{1}{3}} \right)$$

$$(\bar{x}, \bar{y}) = \boxed{\left(\frac{1}{2}, \frac{3}{2} \right)}$$

7. For this problem, let $g(x) = (3+x)^{3/2}$ and consider the Taylor polynomials based at $b = 1$.

(a) (5 points) Find the second Taylor polynomial, $T_2(x)$, for the function g based at $b = 1$.

$$\begin{aligned} g(x) &= (3+x)^{3/2} & g(1) &= 8 \\ g'(x) &= \frac{3}{2}(3+x)^{1/2} & g'(1) &= 3 \\ g''(x) &= \frac{3}{4}(3+x)^{-1/2} & g''(1) &= \frac{3}{8} \end{aligned}$$

$$T_2(x) = 8 + 3(x-1) + \frac{3}{16}(x-1)^2$$

(b) (4 points) Use $T_2(x)$ to approximate the value of $\sqrt{4.4^3}$

$$\sqrt{4.4^3} = (3+1.4)^{3/2} = g(1.4)$$

$$g(1.4) \approx T_2(1.4) = 8 + 3(0.4) + \frac{3}{16}(0.4)^2 = 9.23$$

$$\sqrt{4.4^3} \approx 9.23$$

(c) (5 points) Use Taylor's inequality to find an upper bound for the error in your approximation in part (b).

$$|g'''(x)| = \left| \frac{-3}{8}(3+x)^{-3/2} \right| = \frac{3}{8\sqrt{(3+x)^3}} \quad \begin{array}{l} \text{largest when } x \text{ is smallest} \\ \text{On interval } [1, 1.4], \text{ can use } x=1: \end{array}$$

$$M = \frac{3}{8 \cdot 4^{3/2}} = \frac{3}{64}$$

$$\text{error} \leq \frac{1}{6} \cdot \frac{3}{64} \cdot |1.4-1|^3 = 0.0005$$

Upper bound:

$$0.0005$$

8. For this problem, let $f(x) = \frac{16}{4-x^2} + 6\cos(2x)$

(a) (6 points) Give the Taylor series for f based at $b = 0$ using one sigma sign.

$$\begin{aligned}
 \frac{1}{1-x} &= \sum_{k=0}^{\infty} x^k \\
 \downarrow \div 4 \\
 \frac{1}{4-4x} &= \sum_{k=0}^{\infty} \frac{x^k}{4} \\
 \downarrow x \rightarrow \frac{x^2}{4} \\
 \frac{1}{4-x^2} &= \sum_{k=0}^{\infty} \frac{x^{2k}}{4^{k+1}} \\
 \downarrow \cdot 16 \\
 \frac{16}{4-x^2} &= \sum_{k=0}^{\infty} \frac{x^{2k}}{4^{k-1}}
 \end{aligned}$$

$$\begin{aligned}
 \cos(x) &= \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!} \\
 \downarrow x \rightarrow 2x \\
 \cos(2x) &= \sum_{k=0}^{\infty} \frac{(-1)^k 2^{2k} x^{2k}}{(2k)!} \\
 \downarrow \cdot 6 \\
 6\cos(2x) &= \sum_{k=0}^{\infty} \frac{6(-1)^k 2^{2k} x^{2k}}{(2k)!}
 \end{aligned}$$

Taylor series:

$$\sum_{k=0}^{\infty} \left(\frac{1}{4^{k-1}} + \frac{6(-1)^k 2^{2k}}{(2k)!} \right) x^{2k}$$

(b) (3 points) Find the largest open interval on which this Taylor series converges.

$$\begin{aligned}
 \frac{1}{1-x} &\text{ converges for } -1 < x < 1 \\
 x &\rightarrow x^2/4 \\
 -1 < \frac{x^2}{4} < 1 \\
 \downarrow \\
 -4 < x^2 < 4 &\rightarrow \text{interval: } (-2, 2)
 \end{aligned}$$

(c) (5 points) Find $T_3(x)$, the third Taylor polynomial for f based at $b = 0$, then estimate the value of $\int_0^1 f(x) dx$ by replacing $f(x)$ with $T_3(x)$.

$$\begin{aligned}
 \text{Taylor series from part (a): } & \overbrace{10}^{k=0} - \overbrace{11x^2}^{k=1} + \overbrace{4.25x^4}^{k=2} + \dots \\
 & T_3(x) \text{ (ignore all terms w/ exponent } > 3) \\
 \int_0^1 f(x) dx &\approx \int_0^1 (10 - 11x^2) dx = \left(10x - \frac{11}{3}x^3 \right) \Big|_0^1 = \frac{19}{3}
 \end{aligned}$$

$$\int_0^1 f(x) dx \approx \frac{19}{3}$$